

Final model · Stage 2: fairness checks and reason codes

Validation split only · run on Kaggle (private CPU notebook) · test still sealed

Result: zip_code is dropped. It added only +0.0004 ROC-AUC (rule: keep only if ≥ 0.002), so a location-based input is not worth its fair-lending risk. The final model is now **M3-B-mono-nozip** (146 features, still 0 monotonicity violations, recalibrated). Every applicant now gets up to 4 plain-language reasons; the most common one for declined applicants is **Loan term**.

A. Does the model need zip_code?

The data has no race, sex or age, so geography is the main **proxy** to check: an area's default history can stand in for the people who live there. Three models were trained with the identical recipe (M3's parameters, learning rate 0.02, monotonic constraints, early stopping):

Model (same recipe, trained on Kaggle)	Features	ROC-AUC	KS	Brier	Rounds
full (147 features)	147	0.7354	0.3400	0.14277	1853
without zip_code	146	0.7351	0.3404	0.14284	1931
without zip_code and state	99	0.7330	0.3365	0.14322	1513

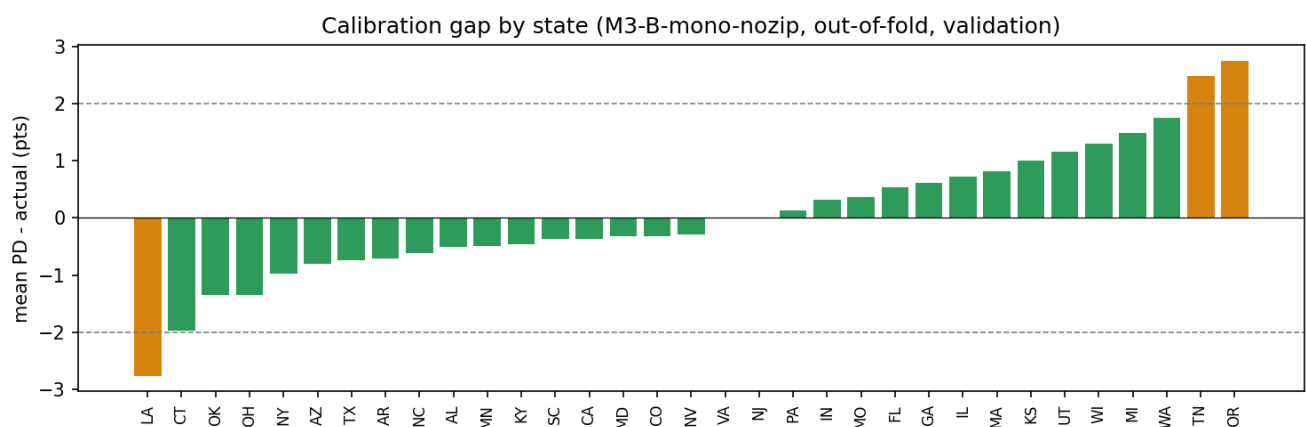
zip_code adds **+0.0004** ROC-AUC; all geography (zip + state) adds +0.0024. Removing zip_code changes the decision for **4,930** applicants (3.0%) under a reference policy that approves the safest 70%:

Area group (by area's historical default rate)	Loans	Actual default rate	Approved with zip	Approved without zip	Change
1 lowest-risk areas	34,182	17.0%	78.2%	75.4%	-2.8 pts
2	31,645	19.6%	72.1%	72.3%	+0.2 pts
3	32,565	20.6%	69.6%	70.1%	+0.5 pts
4	31,853	21.5%	67.0%	67.1%	+0.1 pts
5 highest-risk areas	32,500	23.4%	62.7%	64.9%	+2.2 pts

Read: without zip_code, approvals shift toward the highest-default areas (+2.2 pts) and away from the lowest (-2.8 pts). With it, applicants are partly judged by *where they live*.

B. Is the model equally accurate across states?

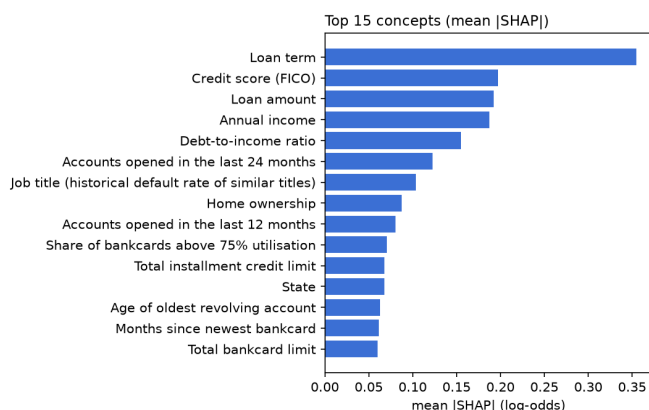
Fairness criterion: **the same PD should mean the same risk everywhere**. For the 33 states with $\geq 1,000$ validation loans (95% of all loans), the calibration gap (mean PD – actual default rate, out-of-fold) ranges from -2.8 to +2.8 points; ROC-AUC from 0.690 to 0.751.



Flagged beyond ± 2 points: **LA, TN, OR**. A positive gap means the state's applicants are charged a higher PD than their actual risk; these states are candidates for monitoring once the model is live.

C. Reason codes: why did this applicant get this PD?

SHAP values split every applicant's score into contributions per input; they add up exactly to the model's output (checked for all 162,745 loans). The 146 columns are grouped into concepts (all `addr_state_*` dummies → "State", missing flags → their feature), and each concept becomes a sentence with the applicant's own value. Up to 4 risk-raising reasons are returned: the usual number of principal reasons in US adverse-action (decline) notices.



Reason (concept)	Share of declined applicants with it in their top 4
Loan term	59.6%
Credit score (FICO)	55.8%
Debt-to-income ratio	43.3%
Annual income	35.7%
Accounts opened in the last 24 months	34.4%
Loan amount	33.7%
Job title (historical default rate of similar titles)	28.2%
Accounts opened in the last 12 months	18.0%
Age of oldest revolving account	10.1%
Total bankcard limit	8.7%

Worked examples (validation applicants; ▲ raised the risk, ▼ lowered it)

Low risk: PD 5.5% (actually repaid) ▲ Loan amount: \$24,000 ▲ Months since last delinquency: 9 months ▲ Delinquencies in the last 2 years: 3 ▲ Months since last bankcard delinquency: 9 months ▼ Annual income: \$160,000 ▼ Loan term: 36 months	Medium risk: PD 16.9% (actually repaid) ▲ Annual income: \$40,587 ▲ Debt-to-income ratio: 32.8% ▲ State: TN ▲ Revolving credit utilisation: 91.5% ▼ Loan term: 36 months ▼ Accounts opened in the last 24 months: 2	High risk: PD 49.1% (actually defaulted) ▲ Loan term: 60 months ▲ Debt-to-income ratio: 37.0% ▲ Annual income: \$28,800 ▲ Accounts opened in the last 12 months: 4 ▼ Share of accounts never delinquent: 75.9% ▼ Charge-offs in the last 12 months: 1
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What we learned

- **Final model after Stage 2:** M3-B-mono-nozip, 146 features, monotonicity violations 0, out-of-fold ROC-AUC 0.7351, Brier 0.14244, mean PD 20.4% vs actual 20.4%.
- Reasons are concrete and checkable by the applicant ("Loan term: 60 months", "Credit score (FICO): 660"), not model jargon.
- Target-encoded inputs such as job title are explained as "historical default rate of similar titles"; like zip codes, job title can be a weak proxy for protected groups and is a candidate for the same ablation test later.
- Limits: fairness here is checked only through geography because the data has no protected attributes; a real lender would also run disparate-impact tests on demographic data it holds separately.

Next: Stage 3, the decision layer (risk bands, approve / review / decline thresholds, expected loss).